The regression equation of X on Y,

$$\hat{X} - 7.875 = 0.169\,(y - 27.5)$$

$$\hat{X} - 7.875 = 0.169y - 0.169 \times 27.5$$

$$\hat{X} = 0.169y + 3.2275$$

Example 5.5

The random sample of 5 school students is selected and their marks in statistics and accountancy are found to be

Statistics	85	60	73	40	90
Accountancy	93	75	65	50	80

Find the two regression lines.

Solution:

The two regression lines are:

Regression equation of Y on X,

$$\hat{Y} - \bar{y} = b_{YX}\left(x - \bar{x}\right)$$

Regression equation of X on Y,

$$\hat{X} - \bar{x} = b_{XY}\left(y - \bar{y}\right)$$

	x_i	y_i	$u_i = x_i - A$ $= x_i - 60$	$v_i = x_i - B$ $= x_i - 75$	$u_i v_i$	u_i^2	y_i^2
	85	93	25	18	450	625	324
	60 A	75 B	0	0	0	0	0
	73	65	13	−10	−130	169	100
	40	50	−20	−25	500	400	625
	90	80	30	5	150	900	25
Total	348	363	48	12	970	2094	1074

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{348}{5} = 69.6$$

$$\bar{y} = \frac{\sum_{i=1}^{n} y_i}{n} = \frac{363}{5} = 72.6$$

Since the mean values are in decimals format not as integers and numbers are big, we take origins for x and y and then solve the problem.

12th_Statistics_EM_Unit_5.indd 142 1/28/2020 6:36:40 PM